Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 1 of 8 · Question 8 part whole · 8 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

8	$y = e^{3x}\sin 2x \quad \frac{dy}{dx} = 2e^{3x}\cos 2x + 3e^{3x}\sin 2x$	M1A1
	$\frac{d^2y}{dx^2} = \left(-4e^{3x}\sin 2x + 6e^{3x}\cos 2x\right) + \left(6e^{3x}\cos 2x + 9e^{3x}\sin 2x\right)$	M1A1A1
	$= 12e^{3x}\cos 2x + 5e^{3x}\sin 2x$	
	$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y$	
	$= 12e^{3x}\cos 2x + 5e^{3x}\sin 2x - 6\left(2e^{3x}\cos 2x + 3e^{3x}\sin 2x\right) + 13e^{3x}\sin 2x$	dM1
	$= 12e^{3x}\cos 2x + 5e^{3x}\sin 2x - 12e^{3x}\cos 2x - 18e^{3x}\sin 2x + 13e^{3x}\sin 2x$	ddM1
	$= 0$ *	A1cso **[8]**
ALT	$\frac{dy}{dx} = 2e^{3x}\cos 2x + 3e^{3x}\sin 2x$	M1A1
	$\frac{dy}{dx} = 2e^{3x}\cos 2x + 3y$	
	$\frac{d^2y}{dx^2} = \left(-4e^{3x}\sin 2x + 6e^{3x}\cos 2x\right) + 3\frac{dy}{dx}$	M1A1A1
	$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = \left(-4e^{3x}\sin 2x + 6e^{3x}\cos 2x + 3\frac{dy}{dx}\right) - 6\frac{dy}{dx} + 13y$	dM1
	$= -13y + 6e^{3x}\cos 2x + 9e^{3x}\sin 2x - 3\frac{dy}{dx} + 13y$	
	$= -13y + 3\frac{dy}{dx} - 3\frac{dy}{dx} + 13y = 0$ *	ddM1A1cso **[8]**
M1	Attempt the product rule. 2 terms of the form $\pm ke^{3x}\cos 2x$ and $\pm le^{3x}\sin 2x$ with $k = 1$ or 2 and $l = 1$ or 3	
A1	Fully correct first derivative	
M1	Attempt the second derivative using the product rule *correctly* on either term. Must have at least one of the terms in the first derivative fully correct.	
A1 A1	A1 for each fully correct bracket	
dM1	Substitute their derivatives and y in $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y$ Depends on both previous M marks	
	This and the following M mark may be awarded together.	
ddM1	Remove the brackets	
A1cso	Reach "0" from fully correct work.	
ALT M1A1	As above	
M1	Replace sin term with a y term and attempt the second derivative using the product rule on first term.	
A1A1	A1 Correct bracket A1 Correct second term	
dM1	As above	
ddM1	Obtain an expression which is either all derivatives plus y terms or all trig terms	
A1cso	Reach "0" from fully correct work	

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Answer 1 of 8 · Question 8 part whole · 8 marks

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Question Number	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 2 of 8 · Question 9 part whole · 8 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

9	$u = e^{-t}$ $\quad v = \sin 2t$ $u' = -e^{-t}$ $\quad v' = 2\cos 2t$	
	$\dfrac{dx}{dt} = 2e^{-t}\cos 2t - e^{-t}\sin 2t = 2e^{-t}\cos 2t - x$	M1 A1 A1
	$u = 2e^{-t}$ $\quad v = \cos 2t$ $u' = -2e^{-t}$ $\quad v' = -2\sin 2t$	
	$\dfrac{d^2x}{dt^2} = -4e^{-t}\sin 2t - 2e^{-t}\cos 2t - \dfrac{dx}{dt}$	M1 A1
	$\dfrac{d^2x}{dt^2} = -4x - \left(\dfrac{dx}{dt} + x\right) - \dfrac{dx}{dt} = -5x - 2\dfrac{dx}{dt}$	dM1 dM1
	$\therefore \dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + 5x = 0$ *	A1 cso
		[8]

	Notes
M1	For an attempt to differentiate using the product rule. Must have 2 terms added together.
A1	For one correct term
A1	For two correct terms
M1	Attempts to differentiate $\dfrac{dx}{dt}$
A1	For $\dfrac{d^2x}{dt^2} = -4e^{-t}\sin 2t - 2e^{-t}\cos 2t - \dfrac{dx}{dt}$ oe
dM1	Dependant on previous M mark - for substitution of x and $\dfrac{dx}{dt}$ into $\dfrac{d^2x}{dt^2}$ or for substitution of x, $\dfrac{dx}{dt}$ and $\dfrac{d^2x}{dt^2}$ into the given equation
dM1	Dependant on previous M mark - for simplifying to $\dfrac{d^2x}{dt^2} = -5x - 2\dfrac{dx}{dt}$ oe or All 5 correct terms seen and an attempt to simplify (5 correct terms may be implied by 7 correct terms)
A1 cso	Obtains the given equation or clear working to show that the equation = 0

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 2 of 8 · Question 9 part whole · 8 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 3 of 8 · Question 3 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part whole in this mark scheme extract

3	$\frac{dy}{dx} = 3e^{3x}\sin 2x + 2e^{3x}\cos 2x$	M1A1A1
	Method A $\frac{d^2y}{dx^2} = 3\left(3e^{3x}\sin 2x + 2e^{3x}\cos 2x\right) + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4y$ $2e^{3x}\cos 2x = \frac{dy}{dx} - 3y \Rightarrow 6e^{3x}\cos 2x = 3\frac{dy}{dx} - 9y$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 3\frac{dy}{dx} - 9y - 4y \Rightarrow 13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$	M1 M1 M1 M1A1 [8]
	Method B $\frac{dy}{dx} = 3y + 2e^{3x}\cos 2x$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x, \quad \frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4y$ $2e^{3x}\cos 2x = \frac{dy}{dx} - 3y \Rightarrow 6e^{3x}\cos 2x = 3\frac{dy}{dx} - 9y$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 3\frac{dy}{dx} - 9y - 4y \Rightarrow 13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$	[M1,M1 M1 M1A1]
	Method C $\frac{d^2y}{dx^2} = 3\left(3e^{3x}\sin 2x + 2e^{3x}\cos 2x\right) + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x$ $13y + \frac{d^2y}{dx^2} = 13e^{3x}\sin 2x + (5e^{3x}\sin 2x + 12e^{3x}\cos 2x)$ $6\frac{dy}{dx} = 6(3e^{3x}\sin 2x + 2e^{3x}\cos 2x)$ $(= 18e^{3x}\sin 2x + 12e^{3x}\cos 2x)$ $LHS = 13e^{3x}\sin 2x + (5e^{3x}\sin 2x + 12e^{3x}\cos 2x)$ $= 18e^{3x}\sin 2x + 12e^{3x}\cos 2x$ $RHS = 6(3e^{3x}\sin 2x + 2e^{3x}\cos 2x)$ $= 18e^{3x}\sin 2x + 12e^{3x}\cos 2x$ $LHS = RHS$	[M1 M1 M1 M1A1]
		Total 8 marks

Question	Notes	Marks
3	$y = e^{3x}\sin 2x$	
	For an attempt to differentiate the given expression.	

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 3 of 8 · Question 3 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part whole in this mark scheme extract

	• There needs to be an acceptable attempt to differentiate both terms. $e^{3x}\sin 2x \rightarrow ke^{3x}\sin 2x + le^{3x}\cos 2x$ with $k, l \neq 0$ • There need to be two terms added. $\frac{dy}{dx} = 3e^{3x}\sin 2x + 2e^{3x}\cos 2x$	M1
	One term completely correct. $\frac{dy}{dx} = 3e^{3x}\sin 2x + '2e^{3x}\cos 2x'$ or $\frac{dy}{dx} = '3e^{3x}\sin 2x' + 2e^{3x}\cos 2x$	A1
	Fully correct differentiated expression. $\frac{dy}{dx} = 3e^{3x}\sin 2x + 2e^{3x}\cos 2x$	A1
	Method A	
	For an attempt to find $\frac{d^2y}{dx^2}$ Minimally acceptable attempt is $ke^{3x}\sin 2x + le^{3x}\cos 2x \rightarrow k(me^{3x}\sin 2x + ne^{3x}\cos 2x) + l(pe^{3x}\cos 2x + qe^{3x}\sin 2x)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dx^2} = 3\left(3e^{3x}\sin 2x + 2e^{3x}\cos 2x\right) + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x$	M1
	For substituting y and $\frac{dy}{dx}$ into their $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4y$	M1
	For preparing to eliminate cos2x by rearranging their $\frac{dy}{dx}$ $2e^{3x}\cos 2x = \frac{dy}{dx} - 3y \Rightarrow 6e^{3x}\cos 2x = 3\frac{dy}{dx} - 9y$ Allow errors in arithmetic but not mathematically incorrect process.	M1
	For an unsimplified expression only in terms of y, $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 3\frac{dy}{dx} - 9y - 4y$	M1
	For the required simplified expression with fully correct working. $13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$	A1 [8]
	Method B	
	For an attempt to find $\frac{d^2y}{dx^2}$ using $\frac{dy}{dx} = 3y + 2e^{3x}\cos 2x$	

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 3 of 8 · Question 3 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part whole in this mark scheme extract

	Minimally acceptable attempt is $\frac{dy}{dx} = ky + le^{3x}\cos 2x \rightarrow$ $\frac{d^2y}{dx^2} = k\frac{dy}{dx} + l(pe^{3x}\cos 2x + qe^{3x}\sin 2x)$ k, l as in their first derivative, $p, q \neq 0$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x$, $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 6e^{3x}\cos 2x - 4y$	M1,M1
	For preparing to eliminate cos2x by rearranging their $\frac{dy}{dx}$ $2e^{3x}\cos 2x = \frac{dy}{dx} - 3y \Rightarrow 6e^{3x}\cos 2x = 3\frac{dy}{dx} - 9y$ Allow errors in arithmetic but not mathematically incorrect process.	M1
	For an unsimplified expression only in terms of y, $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 3\frac{dy}{dx} + 3\frac{dy}{dx} - 9y - 4y$	M1
	For the required simplified expression with fully correct working. $13y + \frac{d^2y}{dx^2} = 6\frac{dy}{dx}$	A1
	Method C	
	For an attempt to find $\frac{d^2y}{dx^2}$ Minimally acceptable attempt is $ke^{3x}\sin 2x + le^{3x}\cos 2x \rightarrow$ $k(me^{3x}\sin 2x + ne^{3x}\cos 2x) + l(pe^{3x}\cos 2x + qe^{3x}\sin 2x)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dx^2} = 3\left(3e^{3x}\sin 2x + 2e^{3x}\cos 2x\right) + 6e^{3x}\cos 2x - 4e^{3x}\sin 2x$	M1
	For substituting y and $\frac{d^2y}{dx^2}$ into the LHS of the result.	M1
	For substituting $\frac{dy}{dx}$ into the RHS of the result. For simplifying the LHS of the result and writing as $k(3e^{3x}\sin 2x + 2e^{3x}\cos 2x)$	M1
	For unsimplified expressions for the LHS and RHS of the result in terms of $e^{3x}\sin 2x$ and $e^{3x}\cos 2x$ only and method to simplify both sides	M1
	For correctly simplifying the LHS and RHS and showing equal.	A1
		Total 8 marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 3 of 8 · Question 3 part whole · 8 marks

2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 4 of 8 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

6(a)(i)	$\frac{dy}{dx} = 4e^{2x} + 2(4x-3)e^{2x}$	M1A1A1 (3)
(ii)	$(4x-3)\frac{dy}{dx} = (4x-3)(8x-2)e^{2x} = (8x-2)y$ *	M1A1cso (2)
(b)	$\frac{dy}{dx} = \frac{5\cos 5x \times (x-3)^2 - \sin 5x \times 2(x-3)}{(x-3)^4}$	M1A1A1 (3)
ALT	Using product rule:	
	$y = (x-3)^{-2}\sin 5x$	M1
	$\frac{dy}{dx} = -2(x-3)^{-3}\sin 5x + 5(x-3)^{-2}\cos 5x$	A1A1
		[8]

(a)(i)

M1 Use product rule to differentiate the given expression. Must have 2 terms added. One to be of the form ke^{2x} and the other of the form $k'(4x-3)e^{2x}$ where $k' = 1$ or 2

A1 Either term correct

A1 Second term correct

NB No simplification needed for these 3 marks

ALT $y = 4xe^{2x} - 3e^{2x} \Rightarrow \frac{dy}{dx} = 4e^{2x} + 8xe^{2x} - 6e^{2x}$

M1 Expand the given expression and differentiate using the product rule for $4xe^{2x}$

A1 Any 2 terms correct; A1 Third term correct.

NB No simplification needed for these 3 marks

(ii)

M1 Use their result from (i) to obtain an expression for $(4x-3)\frac{dy}{dx}$. No need to simplify.

A1cso Correct **given** result obtained with no errors in the working.
Can start with LHS and show equal to the RHS or vice versa or can start with each side and "meet in the middle"

(b)

M1 Attempt the quotient rule. The denominator must be $(x-3)^4$ and the numerator must be of the form $\left(k\cos 5x \times (x-3)^2 - \sin 5x \times l(x-3)\right)$ $k = \pm 5$ or ± 1, $l = 1$ or 2

(ie sine may have been differentiated to − cosine)

A1 One fully correct term in numerator.

A1 All fully correct.

ALT

M1 Rewrite without a quotient and apply the product rule obtaining 2 terms of the form shown

A1 Either term correct

A1 Second term correct

No need to simplify

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 4 of 8 · Question 6 part a.ii · 2 marks

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Question Number	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 5 of 8 · Question 7 part b · 5 marks

2023 JAN · Paper 02 · find part b in this mark scheme extract

7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 6 of 8 · Question 9 part b · 5 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

9 (a)	$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ *	M1A1A1 Cso [3]
(b)	**Method A** $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$ $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y \Rightarrow \frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$ $\Rightarrow \frac{d^2y}{dt^2} = -8\frac{dy}{dt} - 20y \Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 20y = 0$ $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0 \Rightarrow M = 8, N = 20$	M1 M1 M1 M1 A1 [5]
	Method B $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ $\left(= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t\right)$ $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$ $= (16 - 2M)e^{-4t}\sin 2t + (12 - 4M + N)e^{-4t}\cos 2t$ $16 - 2M = 0$ and $12 - 4M + N = 0$ $M = 8,\ N = 20$	[M1 M1 M1 M1 A1]
	Method C By inspection – please see additional guidance.	
		Total 8 marks

Part	Mark	Additional Guidance

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Answer 6 of 8 · Question 9 part b · 5 marks

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(a)	M1	For an attempt to differentiate the given expression. • There needs to be an acceptable attempt to differentiate both terms. $e^{-4t}\cos 2t \rightarrow ke^{-4t}\sin 2t + le^{-4t}\cos 2t$ with $k, l \neq 0$ • There need to be two terms added. $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t$
	A1	One term completely correct. $\frac{dy}{dt} = -4e^{-4t}\cos 2t + '-2e^{-4t}\sin 2t$ OR $\frac{dy}{dt} = '-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t$
	A1	Fully correct differentiated expression and rearranged into the required form $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ Accept also, For example: $-\left(-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t\right) - 4\left(e^{-4t}\sin 2t\right) = 2e^{-4t}\sin 2t$ With any unambiguous indication that a conclusion has been reached, e.g., QED, hash sign, double underline, RHS = LHS etc.
		Method A
(b)	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4x}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y and $\frac{dy}{dt}$ into their $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$
	M1	For preparing to eliminate $\cos 2t$ by rearranging their $\frac{dy}{dt}$ $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ Allow errors in arithmetic but not mathematically incorrect process.
	M1	For an unsimplified expression only in terms of y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$
	A1 [8]	For finding the value of M and the value of N. $M = 8,\ N = 20$
		Method B

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Answer 6 of 8 · Question 9 part b · 5 marks

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	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \to k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ into the equation. $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$
	M1	For collecting terms $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = ('16' - 2M)e^{-4t}\sin 2t + ('12' - 4M + N)e^{-4t}\cos 2t$
	M1	For identifying two equations from their working and a method to solve them simultaneously For example, $'16' - 2M = 0$ and $'12' - 4M + N = 0$
	A1	For finding the value of M and the value of N. $M = 8, N = 20$
		Method C (By inspection)
	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \to k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For simplifying $\frac{d^2y}{dt^2} = e^{-4t}16\sin 2t + e^{-4t}12\cos 2t = \left[e^{-4t}(16\sin 2t + 12\cos 2t)\right]$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}[\sin 2t]$ and deducing that $'16' + (\underline{8} \times -2) = 0 \Rightarrow M = 8$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ and y and deducing that $'12' - ('8' \times 4) + \underline{20} = 0 \Rightarrow N = 20$
	A1	For both $M = 8$ and $N = 20$
		Total 8 marks

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Answer 6 of 8 · Question 9 part b · 5 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 7 of 8 · Question 6 part whole · 8 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

6	$\frac{dy}{dx} = 4e^{4x}\cos 3x - 3e^{4x}\sin 3x$ $\left[\frac{dy}{dx} = 4y - 3e^{4x}\sin 3x\right]$ $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 3\left[4e^{4x}\sin 3x + 3e^{4x}\cos 3x\right]$ $\left\{12e^{4x}\sin 3x = 16y - 4\frac{dy}{dx}\right\}$ $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 16y + 4\frac{dy}{dx} - 9y \Rightarrow \frac{d^2y}{dx^2} + 25y = 8\frac{dy}{dx}$ $A = 25 \quad B = 8$	M1A1A1 M1A1 M1 M1A1cso [8]
ALT	$\frac{dy}{dx} = 4e^{4x}\cos 3x - 3e^{4x}\sin 3x$ $\frac{d^2y}{dx^2} = 16e^{4x}\cos 3x - 12e^{4x}\sin 3x - 12e^{4x}\sin 3x - 9e^{4x}\cos 3x$ $= 7e^{4x}\cos 3x - 24e^{4x}\sin 3x$ Substituting into given equation: $\frac{d^2y}{dx^2} + Ay = B\frac{dy}{dx} \Rightarrow$ $7e^{4x}\cos 3x - 24e^{4x}\sin 3x + Ae^{4x}\cos 3x = B\left(4e^{4x}\cos 3x - 3e^{4x}\sin 3x\right)$ Equating coefficients $7 + A = 4B \qquad -24 = -3B \Rightarrow B = 8$ $7 + A = 4 \times 8 = 32$ $\Rightarrow A = 25$	M1A1A1 M1A1 M1 M1 A1 [8]
		Total 8 marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 7 of 8 · Question 6 part whole · 8 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
6 M1	For an attempt at product rule: • The formula used must be correct • $e^{4x} \to ke^{4x} \quad k \neq 0,1$ • $\cos 3x \to -l\sin 3x \quad l \neq 0,1$ • Minimally acceptable attempt is: $ke^{4x}\cos 3x - le^{4x}\sin x$
A1	At least one term must be correct
A1	Fully correct. Ignore poor notation for $\frac{dy}{dx}$
M1	For an attempt at the second derivative.
A1	For the correct second derivative - simplified or unsimplified.
M1	For converting $me^{4x}\sin 3x$ into y's and $\frac{dy}{dx}$'s i.e., $-me^{4x}\sin 3x = ny - p\frac{dy}{dx}$
M1	For changing the expression in terms of $y, \frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ only
A1cso	For the correct expression exactly as given in the question.
ALT	
M1	For an attempt at product rule: • The formula used must be correct • $e^{4x} \to ke^{4x} \quad k \neq 0,1$ • $\cos 3x \to -l\sin 3x \quad l \neq 0,1$ • Minimally acceptable attempt is: $ke^{4x}\cos 3x - le^{4x}\sin x$
A1	At least one term must be correct
A1	Fully correct. Ignore poor notation for $\frac{dy}{dx}$
M1	For attempting the second derivative. Ft their $\frac{dy}{dx}$ and look for exactly 4 terms, with an attempt to differentiate as for the first M1
A1	Fully correct second derivative – simplified or unsimplified. If it correct unsimplified, then isw and award this mark.
M1	For setting up the equation with A and B as shown in the main scheme.
M1	Equating **their** coefficients to produce two equations in A and B
A1	For the correct values of A and B or embedded in the equation $\frac{d^2y}{dx^2} + 25y = 8\frac{dy}{dx}$

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 7 of 8 · Question 6 part whole · 8 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 8 of 8 · Question 5 part whole · 7 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

5	$\frac{dy}{dx} = 2e^{2x}\left(x^2 - 5x\right) + e^{2x}\left(2x - 5\right) \Rightarrow \left[e^{2x}\left(2x-5\right) = \frac{dy}{dx} - 2y\right]$	M1A1A1
	$\frac{d^2y}{dx^2} = 4e^{2x}\left(x^2 - 5x\right) + 2e^{2x}\left(2x-5\right) + 2e^{2x}\left(2x-5\right) + 2e^{2x}$	M1A1
	$\left[\text{OR } \frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2e^{2x}\left(2x-5\right) + 2e^{2x}\right]$	
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left[\frac{dy}{dx} - 2y\right] + 2e^{2x}$	M1
	$\Rightarrow 2e^{2x} = \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y \quad *$	A1 cso [7]
	ALT 1 LHS = RHS	
	$\frac{dy}{dx} = 2e^{2x}\left(x^2 - 5x\right) + e^{2x}\left(2x-5\right) = 2e^{2x}x^2 - 10xe^{2x} + 2e^{2x}x - 5e^{2x}$	
	$\left[= 2e^{2x}x^2 - 8e^{2x}x - 5e^{2x}\right]$	[M1A1A1
	$\frac{d^2y}{dx^2} = 4e^{2x}x^2 + 4e^{2x}x - 8e^{2x} - 16e^{2x}x - 10e^{2x}$	
	$\left[= 4e^{2x}x^2 - 12e^{2x}x - 18e^{2x}\right]$	M1A1
	$\mathbf{RHS} = 4e^{2x}x^2 - 12e^{2x}x - 18e^{2x} - 4\left(2e^{2x}x^2 - 8e^{2x}x - 5e^{2x}\right) + 4\left(e^{2x}x^2 - 5e^{2x}\right)$	
	$\mathbf{RHS} = 4e^{2x}x^2 - 12e^{2x}x - 18e^{2x} - 8e^{2x}x^2 + 32e^{2x}x + 20e^{2x} + 4e^{2x}x^2 - 20e^{2x}x$	M1
	$\mathbf{RHS} = 2e^{2x} = \mathbf{LHS} \quad *$	A1]
	ALT 2 LHS = RHS	
	$\frac{dy}{dx} = 2e^{2x}\left(x^2 - 5x\right) + e^{2x}\left(2x-5\right) = \left[e^{2x}\left(2x^2 - 8x - 5\right)\right]$	[M1A1A1
	$\frac{d^2y}{dx^2} = e^{2x}\left(4x-8\right) + 2e^{2x}\left(2x^2 - 8x - 5\right) = \left[e^{2x}\left(4x^2 - 12x - 18\right)\right]$	M1A1
	$\mathbf{RHS} = e^{2x}\left(4x^2 - 12x - 18\right) - 4\left(e^{2x}\left(2x^2 - 8x - 5\right)\right) + 4\left(e^{2x}\left(x^2 - 5x\right)\right)$	
	$\mathbf{RHS} = 4e^{2x}x^2 - 12e^{2x}x - 18e^{2x} - 8e^{2x}x^2 + 32e^{2x}x + 20e^{2x} + 4e^{2x}x^2 - 20e^{2x}x$	M1
	$\mathbf{RHS} = 2e^{2x} = \mathbf{LHS} \quad *$	A1]
		Total 7 marks

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 8 of 8 · Question 5 part whole · 7 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
General principles of marking this question. • You must read every line of working carefully. • The first 3 marks are for the correct application of Product Rule • The next 2 marks are for the correct reapplication of Product rule. • The final M mark is effectively for rearranging/substituting to the required form. • The final mark is only to be awarded if everything is correct**. This is a show question.** **Some candidates may use implicit differentiation, this is absolutely fine.**	
M1	For an attempt at product rule. • There must be an acceptable attempt to differentiate both terms. • The correct formula must be applied. • Minimally acceptable differentiation is as follows: $e^{2x} \Rightarrow \pm ke^{2x}$ and $x^2 - 5x \Rightarrow \pm(2x-5)$
A1	For one term correct within product rule
A1	For a fully correct derivative.
M1	For differentiating their first derivative a second time to find the second derivative. There are several different ways this can be completed. Please apply the same rules for the second derivative as for the first. There must be only 4 terms here out of which, award this mark for at **least two** terms correct.
A1	For a fully correct $\frac{d^2y}{dx^2}$ in any form
M1	For substituting $e^{2x}(2x-5) = \frac{dy}{dx} - 2y$
A1cso	For the correct final expression with no errors.
ALT 1 and 2	
M1	For an attempt at product rule. • There must be an acceptable attempt to differentiate both terms. • The correct formula must be applied. • Minimally acceptable differentiation is as follows: $e^{2x} \Rightarrow \pm ke^{2x}$ and $x^2 - 5x \Rightarrow \pm(2x-5)$
A1	For one term correct within product rule
A1	For both terms fully correct.
M1	For differentiating their first derivative a second time to find the second derivative. There are several different ways this can be completed. Please apply the same rules for the second derivative as for the first. Look for two out of four terms correct If their starting point is $\frac{dy}{dx} = e^{2x}(2x^2 - 8x - 5)$ or $2e^{2x}x^2 - 8e^{2x}x - 5e^{2x}$ check their differentiation for the M mark and apply 1 out of 2, or 2 out of 4 correct..
A1	For a fully correct $\frac{d^2y}{dx^2}$ in any form
M1	Collects up all terms on the RHS to obtain Ke^{2x} $K \neq 0$ **Please check their working carefully.**
A1cso	For the correct final expression with no errors.

Type MS · Differentiating twice and proving a relation between y and its derivatives

Answer 8 of 8 · Question 5 part whole · 7 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks